Name: _____ MTH234 Linear Algebra

ID: _____ 1/2022 (International Section)

MIDTERM EXAM (Total 23 points)

Instruction: Students must provide all details of your works. You may use any class materials. Calculator and other materials are not allowed.

1. Let $B = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}$.

   (a) (2 points) Find B^{-1} if exists.

   (b) (4 points) Find all matrices A such that $AB = BA$.

2. (5 points) Use elementary row operations to find A^{-1} (if any) where

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 2 & 0 \\ 2 & 0 & 5 \end{bmatrix}.$$

3. (6 points) Use Cramer's rule to solve the following system of equation:

$$\begin{aligned} 2x_1 + x_2 - x_3 &= 4 \\ x_1 - 2x_2 + x_3 &= 1 \\ 3x_1 - 3x_2 + 2x_3 &= 0 \end{aligned}$$

4. (6 points) Use elementary row operations to determine whether the following system of linear equation is consistent or not, if it is, find the general solution.

$$\begin{aligned} x_1 - 3x_2 + x_3 &= 2 \\ 3x_1 - 8x_2 + 2x_3 &= 5 \\ 2x_2 - 2x_3 &= -2 \end{aligned}$$